

Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

June 5, 2023

Instructions: Read the questions carefully, show all your work, and explain all your answers. Indicate carefully if you introduce notation, or make assumptions when answering a question. Please make sure to allocate time to ALL three problems.

Good luck!

1. Consider the following overlapping generations economy without capital. Individuals live for two periods. The cohort size is constant over time. In each cohort, there are two types of agents, A and B . There is an equal number of agents of each type.

Individuals receive an endowment when young and old. An individual of type $i \in \{A, B\}$, has an endowment of z^i of the consumption good when young. When old, each individual has an endowment that is a given fraction of their endowment when young. Formally, an individual of type $i \in \{A, B\}$ has an endowment when old equal to $z^i(1 - \delta^i)$, with $\delta^i \in [0, 1]$.

There is a market for consumption loans in each period that pays the (gross) rate of return R_t . Individuals value consumption in each period and discount the future at rate $\beta > 0$. An individual of type $i \in \{A, B\}$ born at time t maximizes

$$U(c_t^i, c_{t+1}^i) = \ln(c_t^i) + \beta \ln(c_{t+1}^i).$$

- (a) Formulate the problem of a young individual born at time t .
- (b) Solve for the equilibrium (constant) interest rate in this economy, and show that it is strictly less than $1/\beta$.
- (c) Show that there is no trade in equilibrium if $\delta^A = \delta^B = \bar{\delta}$. Provide intuition for this result.
- (d) Suppose now that $z_A > z_B$, $\delta^A = \bar{\delta} + \Delta$, and $\delta^B = \bar{\delta} - \Delta$, where $\Delta \in (0, \bar{\delta})$. That is, the values of individual endowments when young are positively associated with the fraction of the endowment loss when old. How does the equilibrium interest rate compare to the one in part (c)? Explain your answer intuitively.

2. Consider the following version of the Lucas tree model. There are two assets: a risky stock and a one-period risk-free bond. The stock's dividend is denoted by z_t . It follows a Markov process, where the distribution of z_{t+1} is given by $F(z_{t+1}|z_t)$. The stock is perfectly divisible and is in fixed supply normalized to one. Purchasing a unit of the risk-free bond this period pays one unit of the consumption good in the next period only. The bond is in zero net supply.

There is a continuum of measure one of identical agents with preferences over consumption and bond holdings given by

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t + \kappa b_t),$$

where c_t is consumption at time t , b_t is the period's t bond holdings (purchased in the previous period, $b_t > 0$ corresponds to saving, $b_t < 0$ corresponds to borrowing), function u is strictly increasing, strictly concave, C^2 , and $\beta \in (0, 1)$. The parameter $\kappa \geq 0$ captures the fact that agents explicitly value holding the risk-free bond. Agents are initially endowed with one unit of the stock and zero bonds.

You are asked to analyze a recursive competitive equilibrium in which the equilibrium price functions are stationary.

- (a) Write down the problem of the representative agent in a recursive form. Derive the stochastic Euler equations for the stock and bond, evaluated at the equilibrium allocations and prices, and interpret them intuitively.
- (b) From now on, assume that the utility function is logarithmic, $u(c) = \ln(c)$. Find the closed-form expression for (i) the equilibrium stock and bond prices; (ii) the expected return on the stock and the return on the risk-free bond (i.e., the risk-free rate).
- (c) Is the equity premium increasing/decreasing/independent of κ ? Explain your answer intuitively.

3. Consider an infinite-horizon discrete-time economy which has access to a linear production technology $f(k) = Ak$, where $A > 0$ and k is the asset used in production. The output produced using this technology can be used for consumption or saved as an asset for the next period. There are two identical agents in this economy whose preferences are given by

$$\sum_{t=0}^{\infty} \beta^t \ln(c_t),$$

where $\beta \in (0, 1)$ is the time discount factor and c_t is an agent's consumption in period t . In period 0, the asset's value is k_0 and the corresponding production of the final good is Ak_0 .

In what follows, you are asked to compare two alternative scenarios of asset management/ownership — equal initial ownership split followed by independent management *versus* joint ownership with alternating management rights.

- (a) Suppose that the asset is equally split between the agents in period 0, and thereafter each agent manages the asset independently (with each agent having access to technology $f(\cdot)$). Set up the agent's problem recursively and derive the saving rule. How does the aggregate stock of asset evolve over time?
- (b) Now suppose that the agents jointly own the asset and have alternating management rights. Namely, in even periods agent 1 (she) decides how much of the final good to save for the next period and how much to withdraw for her own consumption in this and the next period. Similarly, in odd periods, agent 2 (he) withdraws from the joint asset to sustain his consumption in two consecutive periods. The asset that is withdrawn by an agent and not consumed immediately continues growing at rate A (the same as the production technology), but only for one period, after which any unconsumed asset that was withdrawn perishes.

Each agent takes decisions of the other agent as given. That is, when agent 1 decides how much to save in the even periods, she takes agent's 2 saving rule as given, and vice versa. Denote the agent i 's saving rule by $k_{t+1} = g_i(k_t)$, $i = 1, 2$.

Agent i 's decision problem can be written recursively as

$$\begin{aligned} V_i(k) &= \max_{c, c', s, k' \geq 0} \ln(c) + \beta \ln(c') + \beta^2 V_i(g_{-i}(k')) \\ \text{s.t. } c + s + k' &= Ak, \\ c' &= As, \end{aligned}$$

where c and c' are the consumption in the current and the next periods, s is the amount of goods privately stored for the next period, and $g_{-i}(\cdot)$ is the saving rule of the other agent, who will manage the asset in the next period.

Show that there exists a solution in which agents use the same linear saving rules. How does the aggregate stock of asset evolve over time in this equilibrium?

- (c) Under which management scenario does the aggregate stock of asset grow slower? Provide an intuitive explanation for your answer.